

11.6 Absolute Convergence and the Ratio Test

1 Absolute and conditional convergence

Consider these two series, both of which converge:

$$(A1) \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

$$(B1) \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

If we make all of the terms positive (i.e. take the absolute value of the individual terms), we get these two series:

$$(A2) \quad \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$(B2) \quad \sum_{n=1}^{\infty} \frac{1}{n}$$

(A2) still converges, but (B2) now diverges. To describe this behavior, we say that

(A1) is **absolutely convergent** and (B1) is **conditionally convergent**.

1.0.1 Example:

Are these series Absolutely Convergent, Conditionally Convergent, or Divergent?

$$\sum_{n=1}^{\infty} \left(-\frac{2}{5}\right)^{n+1}$$

AC

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}} \text{ C}$$

$$\sum \frac{1}{\sqrt{n}} \text{ D}$$

C.C.

$$\sum_{n=1}^{\infty} \frac{1}{n^3} \text{ C}$$

A.C.

$$\sum_{n=1}^{\infty} \left(-\frac{3}{2}\right)^n$$

D

$$\sum_{n=1}^{\infty} \left(\frac{-1}{n}\right) = -\sum \frac{1}{n}$$

D

$$\sum_{n=1}^{\infty} 4^{-n} = \sum_{n=1}^{\infty} \frac{1}{4^n}$$

AC

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \leftarrow \text{C by AST}$$

C.C.

$$\sum \frac{1}{n} \text{ D.}$$

2 Ratio Test

The Ratio Test

To test the series

$$\sum_{n=1}^{\infty} a_n,$$

take the limit

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L = \lim_{n \rightarrow \infty} |a_{n+1}| \cdot \frac{1}{|a_n|}$$

- If $L < 1$ the series is absolutely convergent.
- If $L > 1$ the series is divergent. (This include if $L = \infty$.)
- If $L = 1$ the ratio test gives no information.

On the remaining examples, determine if the given series is Absolutely Convergent, Conditionally Convergent, or Divergent.

2.0.1 Example:

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n + 4}$$

$$a_n = \frac{n^2}{3^n + 4}$$

$$a_{n+1} = \frac{(n+1)^2}{3^{n+1} + 4}$$

$$\lim_{n \rightarrow \infty} \frac{(n+1)^2}{3^{n+1} + 4} \cdot \frac{3^n + 4}{n^2} = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{n^2} \cdot \frac{3^n + 4}{3^{n+1} + 4}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 + 2n + 1}{n^2} \cdot \lim_{n \rightarrow \infty} \frac{3^n + 4}{3^{n+1} + 4}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2}{n^2} \cdot \lim_{n \rightarrow \infty} \frac{3^n}{3^{n+1}}$$

$$= 1 \cdot \frac{1}{3} = \frac{1}{3} < 1 \quad \therefore \boxed{\text{Converge}}$$

2.0.2 Example:

$$\sum_{n=1}^{\infty} \frac{n^9 + n^2}{n!}$$

$$\lim_{n \rightarrow \infty} \frac{(n+1)^9 + (n+1)^2}{(n+1)!} \cdot \frac{n!}{n^9 + n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)^9 + (n+1)^2}{n^9 + n^2} \cdot \frac{n!}{(n+1)!}$$

$$= \lim_{n \rightarrow \infty} \frac{\boxed{n^9 + \dots + n^2 + 2n + 1}}{\boxed{n^9 + n^2}} \cdot \frac{\cancel{n!}}{(n+1)\cancel{n!}}$$

$$= \lim_{n \rightarrow \infty} \frac{n^9}{n^9} \cdot \lim_{n \rightarrow \infty} \frac{1}{n+1} = 1 \cdot 0 = 0 < 1 \quad \therefore \boxed{AC}$$

2.0.3 Example:

$$\sum_{n=1}^{\infty} \frac{n^2 + n + 1}{2^n}$$

2.0.4 Example:

$$\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{3n+4}{n^2} \right)$$

$$\lim_{n \rightarrow \infty} \frac{3(n+1)+4}{(n+1)^2} \cdot \frac{n^2}{3n+4} = \lim_{n \rightarrow \infty} \frac{3n+3+4}{3n+4} \cdot \frac{n^2}{n^2+2n+1}$$

$| \cdot | = |$ indeterminate

→ AST.

$$\lim_{n \rightarrow \infty} \frac{3n+4}{n^2} = 0 \quad \therefore \text{C. by AST.}$$

2.0.5 Example:

$$\sum_{n=1}^{\infty} (-1)^n \left(\frac{3^n}{n^2 \cdot 2^n} \right)$$

2.0.6 Example:

$$\sum_{n=1}^{\infty} \frac{(-1)^n n^3 \cdot 10^n}{n!}$$